

ASTRA FLIGHT CONTROLLER

Complete Technical Reference Manual

STM32F411CEU6 • ICM-42688-P • USB Type-C • 100 mm × 100 mm PCB

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Chapter 1 – Introduction

1.1 Project Overview

1.1.1 Introduction

The Astra Flight Controller is a custom-designed flight control board developed for multirotor unmanned aerial vehicles (UAVs). It is built around the STM32F411CEU6 microcontroller and the ICM-42688-P six-axis Inertial Measurement Unit (IMU), providing a compact, high-performance platform capable of running modern flight control firmware such as Betaflight, INAV, or custom-developed firmware.

Unlike commercial flight controllers, Astra has been designed from the ground up with a strong focus on learning, customization, and hardware understanding. Every component on the PCB has been selected manually, every connection has been verified, and the entire schematic and PCB layout have been created using KiCad.

This document serves as the complete technical reference for the Astra Flight Controller. It explains the design choices, hardware architecture, power distribution, component selection, PCB layout practices, assembly recommendations, and debugging procedures.

1.1.2 Main Features

The Astra Flight Controller includes the following hardware features:

- STM32F411CEU6 ARM Cortex-M4 Microcontroller
- ICM-42688-P 6-axis IMU
- MP2359DJ High-Efficiency 5 V Buck Converter
- AP2112K 3.3 V Low-Dropout Voltage Regulator
- USB Type-C Interface
- SWD Programming Header
- XT30 Battery Input
- Active Buzzer Driver Circuit
- Status LED
- External UART Expansion
- Dedicated SPI Interface for the IMU
- Compact PCB designed within a 100 mm × 100 mm manufacturing limit

1.1.3 Applications

The Astra Flight Controller is suitable for:

- FPV Racing Drones
- Freestyle Quadcopters
- Educational Robotics
- Autonomous UAV Development
- Research Projects
- Flight Control Algorithm Development
- Embedded Systems Learning

1.2 Design Objectives

The purpose of the Astra Flight Controller project extends beyond simply building another flight controller. The project was undertaken to gain practical experience in professional PCB design and embedded hardware development while producing a fully functional and manufacturable controller.

The primary design objectives were:

Educational Value

To understand every subsystem involved in designing a modern flight controller, including power regulation, sensor integration, high-speed digital communication, PCB layout, and firmware programming.

Reliability

To create a stable hardware platform using industry-proven components capable of operating under vibration, electrical noise, and varying power conditions encountered in UAV applications.

Manufacturability

To ensure the PCB can be fabricated using standard manufacturing services such as JLCPCB without requiring advanced fabrication processes or expensive custom options.

Serviceability

To make debugging and firmware development straightforward by including:

- Standard SWD programming interface
- Clearly labelled connectors
- Modular circuit blocks
- Easy-to-understand schematic organization

Expandability

The design intentionally leaves room for future improvements, including:

- Barometer support
- Magnetometer support
- Blackbox Flash Memory
- On-board OSD
- Bluetooth or Wi-Fi telemetry
- CAN bus peripherals
- Additional UART interfaces

1.2.1 Project Specifications

Parameter	Specification
Project Name	Astra Flight Controller
PCB Software	KiCad 10
PCB Size	$\leq 100 \text{ mm} \times 100 \text{ mm}$
MCU	STM32F411CEU6
IMU	ICM-42688-P
Main Input Voltage	Battery via XT30
Regulated Outputs	5 V, 3.3 V
Programming Interface	SWD
Parameter	Specification

USB Interface	USB Type-C
Intended Manufacturer	JLCPCB

Chapter 2 – System Overview

2.1 Hardware Architecture

2.1.1 Introduction

The Astra Flight Controller is designed around a modular hardware architecture, where each functional block performs a specific task while interacting with the others through clearly defined interfaces. This modular approach simplifies debugging, improves reliability, and allows future upgrades without requiring a complete redesign.

The hardware is divided into seven major subsystems:

- Power Management
- Processing Unit
- Sensor System
- Communication Interfaces
- User Interface
- Programming & Debugging
- External Connections

Each subsystem operates independently while contributing to the overall functionality of the flight controller.

2.1.2 Hardware Modules Power Management

Responsible for converting battery voltage into stable regulated voltages used by the board.

Major Components:

- XT30 Battery Connector
- Resettable Fuse
- MP2359DJ Buck Converter
- AP2112K 3.3V Regulator Filtering Capacitors Outputs:
- +5V Rail
- +3.3V Rail

Processing Module

The processing module is the "brain" of the flight controller. Main Component: STM32F411CEU6.

Responsibilities:

- Reads sensor data
- Executes flight control algorithms
- Controls motors
- Handles USB communication
- Processes receiver input
- Generates PWM/DShot signals
- Controls LEDs and buzzer

Sensor Module

The sensor subsystem provides real-time motion data. Main Sensor: ICM-42688-P.

Measurements: 3-axis Accelerometer, 3-axis Gyroscope. Communication: SPI Interface.

The IMU continuously measures angular velocity and acceleration, enabling the firmware to estimate the drone's orientation and movement.

Communication Module

Communication interfaces allow the controller to interact with external devices. Interfaces include USB Type-C, SWD Programming, SPI, and UART. These interfaces are used for firmware flashing, debugging, telemetry, and configuration.

User Interface

The user interface consists of simple hardware that provides status information and manual interaction. Components: Status LED, Push Button, Active Buzzer. These components assist during programming, calibration, troubleshooting, and normal operation.

External Interfaces

The controller provides several connectors for integration with other systems: Battery Input, ESC Outputs, UART Expansion, SWD Header, USB-C Connector. These connectors enable communication with motors, sensors, programming tools, and external peripherals.

2.2 System Operation

2.2.1 Power-Up Sequence

When battery power is applied, the following sequence occurs:

- Step 1: Battery voltage enters through the XT30 connector.
- Step 2: The resettable fuse protects the board from excessive current.
- Step 3: The MP2359DJ converts battery voltage into a regulated +5V supply.
- Step 4: The AP2112K converts +5V into +3.3V.
- Step 5: The STM32F411CEU6 receives stable power.
- Step 6: The crystal oscillator starts generating the system clock.
- Step 7: The microcontroller boots from Flash memory.
- Step 8: The SPI interface initializes the ICM-42688-P.
- Step 9: Sensors begin transmitting motion data.
- Step 10: The flight controller enters normal operating mode.

2.2.2 Clock Generation

The STM32F411CEU6 uses an external crystal oscillator for accurate timing. The external crystal provides stable CPU clock, accurate USB timing, reliable SPI communication, and consistent sensor sampling. Proper placement of the crystal and its load capacitors is essential for reliable operation.

2.2.3 Design Philosophy

Several engineering principles were followed throughout the design of the Astra Flight Controller:

Simplicity

The schematic is divided into functional blocks that are easy to understand and maintain.

Reliability

All power rails include local decoupling capacitors, short return paths, and proper grounding techniques.

Expandability

Unused GPIO pins and communication interfaces are available for future hardware additions.

Manufacturability

The PCB is designed to be compatible with standard manufacturing services such as JLCPCB while remaining within the 100 mm × 100 mm board size li

Chapter 3 – Power System

3.1 Introduction

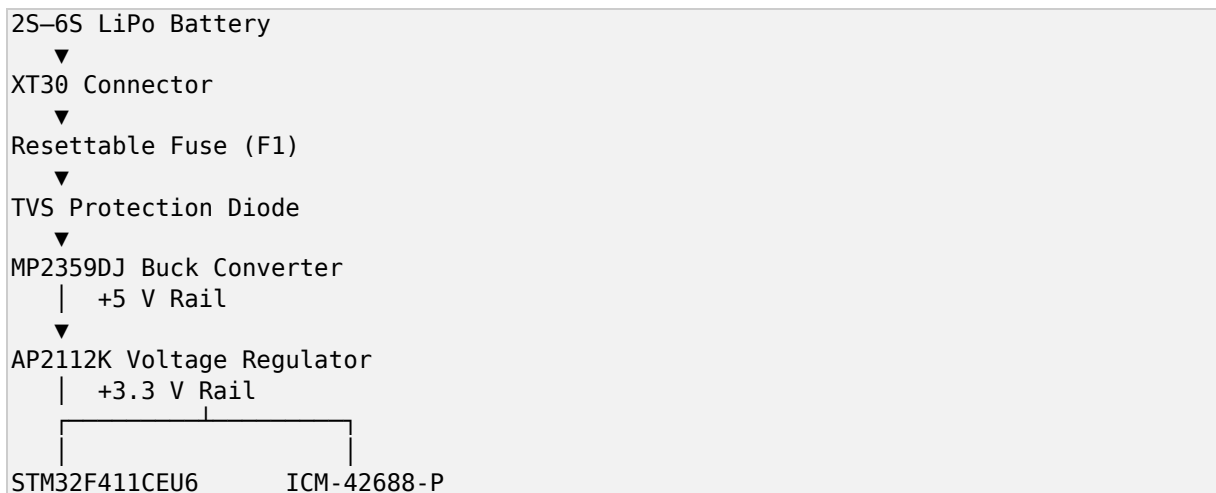
3.1.1 Overview

The power system is one of the most critical sections of the Astra Flight Controller. Every subsystem—including the microcontroller, IMU, USB interface, and peripherals—depends on a stable and clean power supply. A poorly designed power stage can lead to unstable sensor readings, processor resets, communication failures, or complete system failure during flight.

The Astra Flight Controller is designed to accept a 2S–6S LiPo battery as its primary power source. Since the battery voltage varies significantly depending on the number of cells and state of charge, the board incorporates dedicated power conversion and regulation stages to generate stable operating voltages.

3.1.2 Power Architecture

The complete power path of the flight controller is shown below.



The power is converted in two stages: Battery → 5 V using the MP2359DJ switching regulator, then 5 V → 3.3 V using the AP2112K low-dropout regulator. This architecture provides both high efficiency and a clean supply for sensitive digital electronics.

3.1.3 Supported Input Voltage

The controller supports the following battery configurations:

Battery	Nominal Voltage	Fully Charged
2S LiPo	7.4 V	8.4 V
3S LiPo	11.1 V	12.6 V
4S LiPo	14.8 V	16.8 V
5S LiPo	18.5 V	21.0 V
6S LiPo	22.2 V	25.2 V

The MP2359DJ is capable of regulating this wide input range into a stable 5 V

output. **3.2 Power Rails**

The Astra Flight Controller contains three main power domains.

VBAT

VBAT is the raw battery voltage entering the board through the XT30 connector. It is used as the input to the buck converter and is also monitored through a resistor divider connected to the STM32 ADC for battery voltage measurement.

+5 V Rail

The +5 V rail is generated by the MP2359DJ buck converter. This rail supplies the AP2112K voltage regulator, USB-related circuitry, the buzzer, and external peripherals requiring 5 V. Using a switching regulator keeps power loss low even when operating from a 6S battery.

+3.3 V Rail

The +3.3 V rail is generated by the AP2112K low-dropout regulator. It powers the STM32F411CEU6, the ICM42688-P IMU, logic-level circuits, and digital communication interfaces. The AP2112K was selected because of its low output noise and excellent transient response, making it suitable for precision sensor applications.

3.2.1 Why Two Regulators?

Using a single linear regulator directly from the battery would result in excessive heat generation, especially at higher input voltages. Instead, the Astra Flight Controller uses a two-stage regulation approach:

- MP2359DJ Buck Converter — efficiently reduces battery voltage to 5 V.
- AP2112K LDO — produces a clean and stable 3.3 V output with minimal electrical noise. This combination provides both efficiency and stable operation.

3.2.2 Power Design Considerations

Several engineering principles were followed during the design of the power system:

- High-current traces are kept short and wide.
- The buck converter is located close to the battery input.
- Input and output capacitors are placed near the MP2359DJ.
- The AP2112K is positioned close to the STM32 and IMU.
- All IC supply pins include local decoupling capacitors.
- The exposed pad of the STM32 is connected to the ground plane for improved thermal performance and electrical grounding.

Chapter Summary: The Astra Flight Controller uses a robust two-stage power architecture to convert a 2S–6S LiPo battery into regulated 5 V and 3.3 V supplies. The MP2359DJ provides efficient primary voltage conversion, while the AP2112K supplies a clean 3.3 V rail for the microcontroller and IMU. This design balances efficiency, stability, and noise performance, providing a reliable foundation for all other subsystems.

Chapter 4 – STM32F411CEU6 Microcontroller

4.1 Introduction

The STM32F411CEU6 is the primary processing unit of the Astra Flight Controller. It executes the flight control firmware, processes sensor data, generates motor outputs, and manages all communication interfaces.

Parameter	Specification
Manufacturer	STMicroelectronics

Device	STM32F411CEU6
CPU Core	ARM Cortex-M4F
Maximum Clock	100 MHz
Flash	512 KB
SRAM	128 KB
Voltage	1.7–3.6 V
Package	UFQFPN-48 (Exposed Pad)
Debug	SWD

Role

The STM32 reads IMU data through SPI, executes flight control algorithms, generates ESC outputs, monitors battery voltage, controls the LED and buzzer, manages USB communication, and supports firmware programming through SWD.

Power and Clock

The MCU operates from the regulated 3.3 V rail generated by the AP2112K regulator. All VDD pins connect to 3.3 V, all VSS pins and the exposed Pad 49 connect to the ground plane. An external crystal provides an accurate system clock for USB and SPI operation.

Reset and SWD

The NRST pin enables reliable startup and debugging. The SWD interface exposes SWDIO, SWCLK, NRST, 3.3 V and GND for programming with an ST-Link.

GPIO Usage

GPIOs are assigned to SPI, USB, UART, battery sensing, motor outputs, status LED, push button and buzzer, with spare pins reserved for future expansion.

Chapter Summary: The STM32F411CEU6 acts as the processing core of the Astra Flight Controller, coordinating every subsystem while providing the performance and peripheral support required for stable UAV operation.

Chapter 5 – ICM-42688-P Inertial Measurement Unit (IMU)

5.1 Introduction

The ICM-42688-P is the primary motion sensing device used in the Astra Flight Controller. It integrates a 3-axis accelerometer and a 3-axis gyroscope in a compact LGA package. The sensor continuously measures aircraft motion and sends this data to the STM32F411CEU6 through the SPI interface for attitude estimation and flight stabilization.

Why the ICM-42688-P was Selected

The sensor was selected because of its excellent gyroscope noise performance, high sampling rate, low latency, low power consumption, compact size, and broad support in modern flight-control firmware such as Betaflight and INAV.

Main Specifications

Parameter	Specification
Manufacturer	TDK InvenSense
Device	ICM-42688-P
Sensor Type	6-Axis IMU
Accelerometer	3-Axis MEMS
Gyroscope	3-Axis MEMS

Operating Voltage	1.71–3.6 V
Communication	SPI / I ² C
Package	LGA-14 (2.5 × 3.0 mm)
Max SPI Clock	Up to 24 MHz
Operating Temp.	–40°C to +85°C

Working Principle

The accelerometer measures linear acceleration along the X, Y, and Z axes, while the gyroscope measures angular velocity about the same axes. Together they provide six degrees of freedom (6-DoF), allowing the firmware to estimate the aircraft's orientation using sensor-fusion algorithms.

Hardware Integration

The IMU operates from the regulated 3.3 V rail. VDD and VDDIO are connected to 3.3 V, while the GND pins connect directly to the ground plane. Local decoupling capacitors are placed close to the power pins to reduce supply noise and improve stability.

SPI Communication

Communication with the STM32F411CEU6 takes place over SPI using SCK, MOSI, MISO, and CS. SPI was selected instead of I²C because it offers higher bandwidth, lower latency, and improved reliability in electrically noisy environments.

Interrupt Output

The interrupt pin notifies the STM32 whenever new sensor data is available, reducing processor workload and improving the timing of the flight-control loop.

PCB Design Considerations

The IMU is placed close to the centre of the PCB, away from the switching regulator and high-current traces. SPI traces are kept short, routed together, and referenced to a continuous ground plane. High-current routing beneath the IMU is avoided to minimise electrical noise.

System Operation

The IMU continuously measures aircraft motion, sends accelerometer and gyroscope data to the STM32 through SPI, and the firmware processes this information to estimate attitude and generate motor control outputs.

Chapter Summary: *The ICM-42688-P provides accurate motion sensing for the Astra Flight Controller. Proper power supply design, PCB placement, grounding, and SPI routing ensure reliable sensor performance and stable flight operation.*

Chapter 6 – USB Type-C Interface

6.1 Introduction

The USB Type-C interface serves as one of the primary communication and power interfaces of the Astra Flight Controller. It provides firmware flashing, debugging, configuration, diagnostics and development power.

USB Type-C offers reversible insertion, improved durability, better current capability and long-term compatibility. The design uses the USB 2.0 Full-Speed interface connected directly to the STM32F411CEU6.

Purpose

Firmware upload, configuration using Betaflight/INAV/STM32CubeProgrammer, debugging through serial communication, and powering the board during development without a LiPo battery.

Advantages

Reversible connector, stronger mechanical retention, higher current capability and modern industry-standard compatibility.

6.2 USB Connector

Pin	Function
VBUS	+5V
GND	Ground
D+	USB Data Positive
D-	USB Data Negative
CC1	Configuration Channel 1
CC2	Configuration Channel 2
Shield	Connector Shield

USB 3.x SuperSpeed pins are unused.

VBUS and Ground

VBUS powers the board during development while the battery remains the primary source during flight. Ground pins and shield connect to the main ground plane for signal integrity and EMI reduction.

6.3 Data Communication

USB Full-Speed uses the D+ and D- differential pair. Route them together with matched lengths, minimal vias, constant spacing, no sharp bends and a continuous ground plane.

The STM32F411CEU6 supports USB 2.0 Full-Speed (12 Mbps), sufficient for firmware updates and configuration.

6.4 Configuration Channel

CC1 and CC2 determine cable orientation and device role. The Astra Flight Controller uses 5.1 kΩ pull-down resistors on both CC pins to identify itself as a USB device and enable 5V from the host.

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Chapter 7 – External Interfaces and Connectors

7.1 XT30 Battery Connector

The XT30 connector is the primary battery input. Battery voltage enters the board through the XT30 connector, passes through the resettable fuse and protection circuitry, and then feeds the MP2359DJ buck converter. Wide copper traces are used to carry battery current and minimise voltage drop.

7.2 SWD Programming Header

Signal	Purpose
SWDIO	Debug Data
SWCLK	Debug Clock
NRST	Reset
3.3 V	Target Reference
GND	Ground

The SWD interface is used with an ST-Link programmer for firmware uploading, debugging and recovery.

7.3 ESC Connections

Motor control signals generated by the STM32 are routed to the ESC outputs. These connections carry logiclevel control signals only; motor power is supplied separately by the battery through the ESCs.

7.4 UART Expansion

UART headers allow connection of peripherals such as GPS modules, telemetry radios, Bluetooth adapters or other serial devices. Keeping UART interfaces accessible makes future upgrades straightforward.

7.5 User Interface

The board includes a status LED, a push button and an active buzzer. The LED provides visual status indication, the push button can be used for user input or boot functions, and the buzzer provides audible alerts for arming, warnings and fault conditions.

7.6 PCB Design Notes

All connectors are placed near the board edge for easy access. High-current connectors are routed with wide traces, while communication connectors are kept away from switching power circuitry to reduce electrical noise.

Chapter 8 – Indicators and User Interface

8.1 Overview

The Astra Flight Controller includes a status LED, a TL3342 push button and an active buzzer. These devices provide visual, audible and manual interaction during development, testing and flight operation.

8.2 Status LED

The STM32 controls the LED to indicate power, boot status, firmware activity, calibration, arming state and error conditions.

State	Meaning
Solid ON	Board powered
Slow Blink	Initialization
Fast Blink	Firmware activity
Pattern	Warning/Error

8.3 Push Button

The TL3342 tactile switch can be used for boot mode selection, firmware recovery, calibration and user-defined functions through a GPIO input.

8.4 Active Buzzer

The active buzzer provides audible feedback including startup, arming, low-battery and fault indications. It requires only an on/off control signal from the STM32.

8.5 PCB Design Considerations

The LED is placed for visibility, the push button is accessible from the board edge, and the buzzer is positioned to maximise sound output while remaining away from sensitive circuitry.

Chapter 9 – PCB Layout & Design Considerations

9.1 Introduction

9.1.1 Overview

The PCB layout is one of the most critical aspects of the Astra Flight Controller. While the schematic defines how components are electrically connected, the PCB layout determines how reliably the circuit performs in real-world conditions. A poorly routed PCB can introduce electrical noise, voltage drops, communication errors, overheating, and unstable sensor readings, even if the schematic is electrically correct.

The Astra Flight Controller has been designed as a compact 2-layer PCB suitable for fabrication through JLCPCB, while maintaining a board size within the 100 mm × 100 mm manufacturing limit. Throughout the layout process, emphasis was placed on minimizing electrical noise, shortening critical signal paths, and ensuring that sensitive analog components remain isolated from high-current switching circuitry.

The layout was created using KiCad 10, following common PCB design practices recommended by STMicroelectronics, TDK InvenSense, and Monolithic Power Systems.

9.1.2 PCB Design Objectives

The PCB layout was designed to satisfy several engineering objectives:

- Reliable operation under electrical and mechanical stress.
- Low electrical noise around the IMU.
- Efficient power distribution.
- Simple manufacturing using standard PCB fabrication services.
- Easy assembly and inspection.
- Good thermal performance.
- Room for future hardware expansion.

9.2 Board Dimensions

50.5mmX50.5mm are the board dimensions, it is kept compact and efficient

9.3 Component Placement Strategy

9.3.1 Microcontroller Placement

The STM32F411CEU6 is positioned near the center of the PCB. This location minimizes the average distance between the MCU and all connected peripherals, resulting in shorter SPI traces, shorter USB connections, reduced propagation delay, easier routing, and improved signal integrity.

The exposed thermal pad beneath the MCU is connected directly to the ground plane for both thermal dissipation and electrical grounding.

9.3.2 IMU Placement

The ICM-42688-P is the most sensitive component on the board. To maximize measurement accuracy, it is placed close to the center of the PCB, kept away from the MP2359DJ buck converter and inductor, and high-current traces are not routed beneath the sensor. A continuous ground plane is maintained underneath the device, and the sensor orientation matches the coordinate system expected by the firmware.

Proper placement reduces electrical interference and vibration-induced measurement errors.

9.3.3 Power Section Placement

The power conversion circuitry is grouped together near the battery input. This includes:

- XT30 connector.
- Resettable fuse.
- Protection diode.
- MP2359DJ buck converter.
- 4.7 μ H Bourns SRP5030TA-4R7M inductor.
- Input and output capacitors. ● AP2112K voltage regulator.

Keeping these components close together minimizes loop area and improves power conversion efficiency.

9.3.4 USB Interface Placement

The USB Type-C connector is located on the edge of the PCB. This provides easy cable access, improved mechanical support, short USB data traces, and simplified assembly. The CC resistors and supporting passive components are placed close to the connector to reduce routing complexity.

9.4 Power Distribution Network (PDN)

9.4.1 Introduction

A Power Distribution Network (PDN) is responsible for delivering stable voltage to every component on the PCB. Even the best voltage regulator cannot guarantee proper operation if the PCB layout introduces excessive resistance, inductance, or voltage drops.

The PDN was designed with three primary objectives: deliver stable voltages under varying load conditions, minimize voltage ripple and switching noise, and ensure that high-current and low-current circuits do not interfere with one another.

The board contains three primary power rails: VBAT (direct battery voltage from the XT30 connector), +5 V (generated by the MP2359DJ buck converter), and +3.3 V (generated by the AP2112K low-dropout regulator).

9.4.2 High-Current Routing

The battery input and switching regulator carry the highest current on the PCB. Design rules applied include:

- Battery traces are made significantly wider than signal traces.
- The distance between the XT30 connector and the buck converter is minimized.
- Switching current loops are kept as small as possible.
- Copper bottlenecks are avoided.
- Multiple vias are used when transferring high-current paths between PCB layers.

9.4.3 Low-Current Distribution

After regulation, the 5 V and 3.3 V rails supply low-power digital circuits. Separate power branches feed different functional blocks, long power traces are avoided, sensitive analog circuits are isolated from noisy switching nodes, and each integrated circuit receives its supply through short connections.

9.5 Decoupling Capacitors

9.5.1 Purpose

Every digital integrated circuit rapidly switches internal transistors on and off, causing short bursts of current demand. Decoupling capacitors act as local energy reservoirs, supplying these transient current demands instantly, reducing voltage ripple, improving power stability, lowering EMI, and enabling reliable microcontroller operation and cleaner sensor measurements.

9.5.2 Bulk Capacitors

In addition to small ceramic decoupling capacitors, larger bulk capacitors stabilize the supply during sudden current changes, reduce low-frequency voltage ripple, improve regulator transient response, and help absorb energy during startup.

9.6 Thermal Management

9.6.1 Heat Sources

Although the Astra Flight Controller has relatively low power consumption, several components generate heat during operation, including the MP2359DJ buck converter, AP2112K regulator, STM32F411CEU6, and the protection diode under high load conditions.

9.6.2 Copper Area

Copper acts as an effective heat spreader. Large copper pours are used around the buck converter, ground copper is connected to exposed thermal pads where applicable, and high-power components are given adequate surrounding copper.

9.6.3 Thermal Vias

Where appropriate, thermal vias connect copper pours on different PCB layers, improving heat transfer and reducing hotspot temperatures. Although the Astra Flight Controller is a two-layer design, strategically placed vias still contribute to heat distribution.

9.7 Manufacturing Files

9.7.1 Gerber Files

Gerber files describe the copper, silkscreen, solder mask, and board outline required for PCB fabrication.

Layer	Purpose
Top Copper	Top electrical traces
Bottom Copper	Bottom electrical traces
Top Solder Mask	Protects top copper
Bottom Solder Mask	Protects bottom copper
Top Silkscreen	Component markings
Bottom Silkscreen	Bottom markings
Board Outline	PCB profile

9.7.2 Drill File

The drill file specifies the location and diameter of every drilled hole on the PCB, including through-hole component holes, mounting holes, and via holes.

9.7.3 Bill of Materials (BOM)

The Bill of Materials lists every component required to assemble the flight controller, including reference designator, component value, manufacturer part number, package, and quantity.

9.8 Manufacturing with JLCPCB

The Astra Flight Controller has been designed to be compatible with JLCPCB's standard PCB fabrication service, considering board dimensions within 100 mm × 100 mm, standard two-layer PCB construction, surface-mount components with commonly available footprints, adequate spacing between components, standard drill sizes, and standard copper thickness.

9.9 Final Hardware Validation

After receiving the manufactured PCB, the following validation procedure should be followed before applying full power:

- Visually inspect the PCB for fabrication defects.
- Check for solder bridges and missing components.
- Verify continuity between power and ground.
- Apply USB power first before connecting a LiPo battery.
- Measure the 5 V rail.
- Measure the 3.3 V rail.
- Verify that the STM32 powers up correctly.
- Confirm SWD communication using an ST-Link programmer.
- Upload test firmware.
- Verify IMU communication over SPI.
- Test the status LED, push button, and buzzer.
- Finally, connect the battery and verify full system operation.

Performing these checks in order minimizes the risk of damaging components and simplifies troubleshooting if problems are found.

Chapter 10 – Power Supply Design

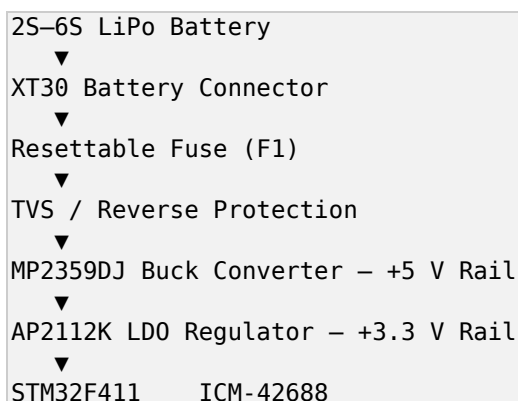
10.1 Introduction

10.1.1 Overview

resets, communication The power supply is the foundation of every electronic system. Regardless of how powerful the processor or how accurate the sensors are, the entire system depends on receiving a clean, stable, and reliable power source. Even small fluctuations in supply voltage can cause unexpected failures, inaccurate sensor readings, or permanent component damage.

The Astra Flight Controller is designed to operate directly from a 2S to 6S Lithium Polymer (LiPo) battery, meaning the input voltage can vary from approximately 7.4 V to 25.2 V. To achieve stable regulation, the Astra Flight Controller uses a two-stage power architecture: the MP2359DJ synchronous buck converter generates a regulated 5 V supply from the battery, and the AP2112K Low Dropout (LDO) regulator generates a clean 3.3 V supply from the 5 V rail.

10.1.2 Power Flow



10.2 Battery Input Stage

10.2.1 XT30 Connector

The flight controller receives power through an XT30 connector, selected for its high current capability, low contact resistance, reliable locking mechanism, wide availability in FPV/UAV applications, and excellent vibration resistance.

10.2.2 Supported Battery Range

Battery	Nominal Voltage	Fully Charged
2S	7.4 V	8.4 V
3S	11.1 V	12.6 V
4S	14.8 V	16.8 V
5S	18.5 V	21.0 V
6S	22.2 V	25.2 V

10.2.3 Resettable Fuse (F1)

Immediately after the XT30 connector, the input passes through a resettable polyfuse (PTC), which protects the

PCB from excessive current caused by short circuits, incorrect wiring, component failures, or assembly mistakes. Unlike a traditional fuse, the polyfuse automatically resets after the fault is removed and the device cools down.

10.2.4 Input Filtering

Before entering the buck converter, the battery supply is filtered using input capacitors placed close to the MP2359DJ. These capacitors reduce voltage ripple, supply instantaneous current during switching, stabilize the converter input, and improve electromagnetic compatibility (EMC).

10.3 MP2359DJ Buck Converter

10.3.1 Why a Buck Converter?

A linear regulator would dissipate the voltage difference between the battery and the output as heat. Converting 25.2 V from a fully charged 6S battery directly to 5 V using a linear regulator would result in excessive power loss and unacceptable temperatures. A switching buck converter solves this problem by rapidly switching the input voltage through an inductor and filtering the output, achieving efficiencies typically above 85–90%.

10.3.2 MP2359DJ Features

- Wide input voltage range.
- High conversion efficiency.
- Integrated MOSFETs.
- Compact package.
- High switching frequency.
- Internal protection features.
- Minimal external components.

10.3.3 Operating Principle

The MP2359DJ is a synchronous step-down (buck) switching regulator. Its operation consists of four basic stages: the internal MOSFET switches ON allowing current to flow into the inductor; energy is stored in the magnetic field of the inductor; the MOSFET switches OFF; and the inductor releases stored energy through the output filter, maintaining a continuous output voltage. This cycle repeats hundreds of thousands of times per second.

10.3.4 External Components

- Input capacitor.
- Output capacitor.
- Bootstrap capacitor.
- Schottky protection diode (where applicable).
- Feedback resistor network.
- 4.7 μ H power inductor.

10.3.5 Selected Inductor

The Astra Flight Controller uses the Bourns SRP5030TA-4R7M, with an inductance of 4.7 μ H, saturation current greater than 3 A, shielded construction, low DC resistance, and a compact SMD package. The shielded design helps reduce electromagnetic interference while providing stable operation at the switching frequency of the regulator.

10.4 Feedback Network

The output voltage of the MP2359DJ is determined by a resistor divider connected between the output voltage and the feedback pin. This network continuously monitors the output voltage and

adjusts the converter duty cycle to maintain a stable 5 V output despite changes in battery voltage or load current.

10.5 AP2112K Low Dropout Regulator

10.5.1 Purpose

Although the 5 V rail generated by the buck converter is highly efficient, switching regulators inherently produce a small amount of high-frequency ripple. The Astra Flight Controller uses an AP2112K LDO to generate a regulated 3.3 V rail, acting as a secondary filtering stage that significantly reduces switching noise before it reaches the microcontroller and IMU.

10.5.2 Advantages of the AP2112K

- Low output noise.
- Good transient response.
- Stable operation with ceramic capacitors.
- Compact footprint.
- Adequate output current for the flight controller.

chapter 11 – Firmware & Programming

11.1 Introduction

11.1.1 Overview

The firmware is the software that transforms the Astra Flight Controller from a collection of electronic components into a fully functional embedded control system. While the hardware provides the computational resources, sensors, and communication interfaces, it is the firmware that coordinates these resources to stabilize the aircraft, interpret pilot commands, and manage peripheral devices.

The firmware runs entirely on the STM32F411CEU6, a 32-bit ARM Cortex-M4F microcontroller. Every operation performed by the flight controller—reading sensor data, calculating aircraft attitude, executing control algorithms, generating ESC outputs, monitoring battery voltage, communicating over USB, and driving the user interface—is controlled by the firmware.

11.1.2 Objectives of the Firmware

- Initialize all onboard hardware correctly during startup.
- Configure the system clock and peripheral interfaces.
- Read sensor data from the IMU.
- Calculate aircraft orientation.
- Execute flight control algorithms.
- Generate motor control outputs.
- Monitor system voltage and health.
- Provide communication through USB and UART.
- Control the LED and buzzer.
- Detect and recover from fault conditions.